

PROCESS WORK – ASSIGNMENT 4 – GAME DESIGN FUNDAMENTALS

Brainstorming

NAMES

- ☒ The Wrath of Raph
- ☐ Raph's Rampage
- ☐ College of the damned
- ☐ Into the fire
- ☐

GAME FACTORS

- Top Down Zelda / resident evil inspired game
- Map of Mohawk College
 - Split up into small pisces
- Player count
 - ☒ 1
 - ☐ 1-2
 - ☐ 1-3
 - ☐ 1-4

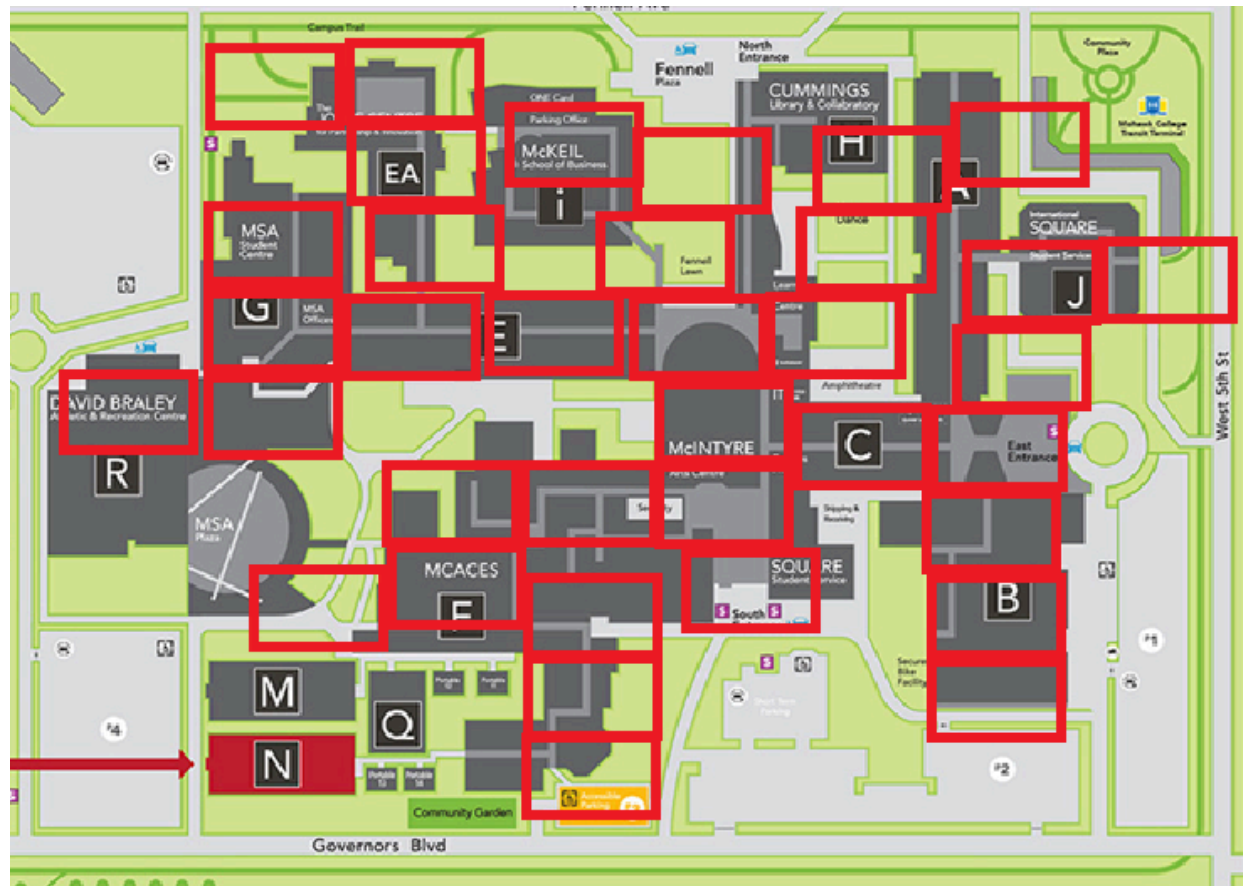
THEME

- ☒ Zombie Apocalypse
- ☐ Alien Invasion ?
- ☐ Nightmare Monsters?

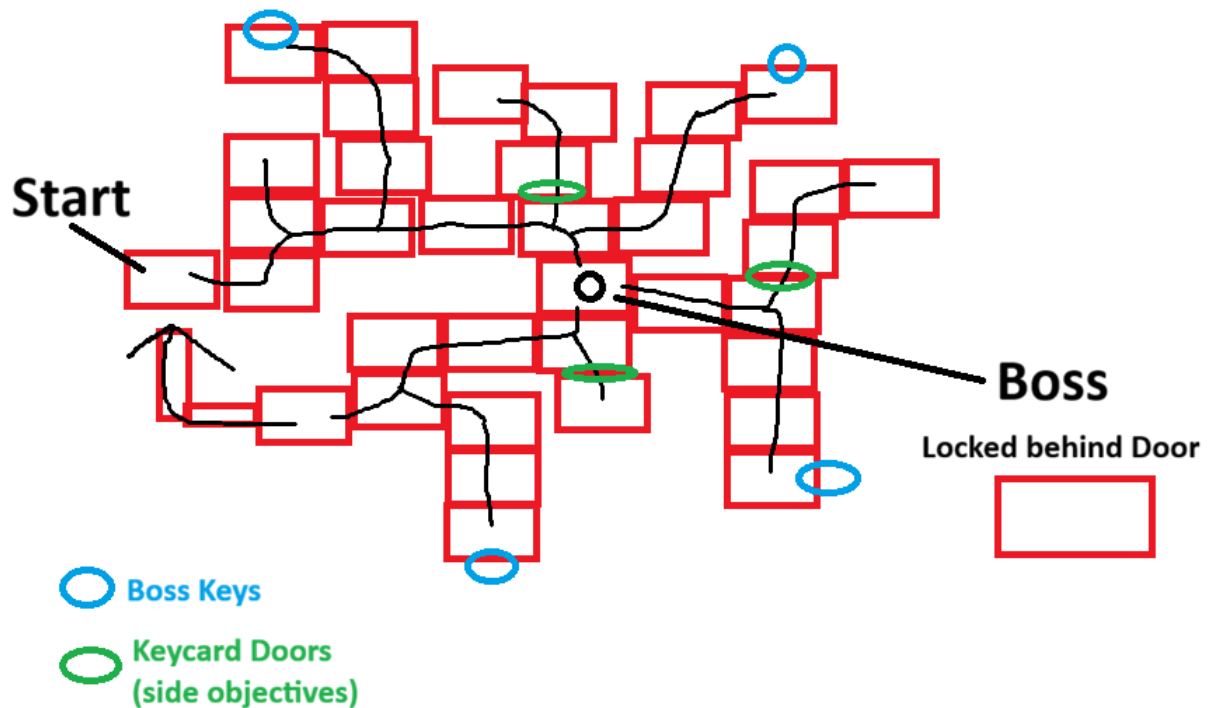
PROJECTILES/ATTACKS

Either Bright and on lower visual layer, or dark and on top layer.

Basic Map Concept (Payton):

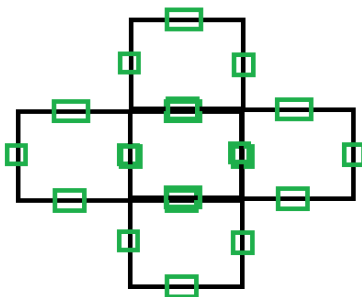
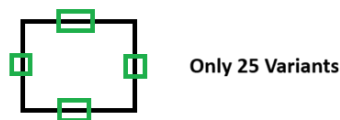


Map Version 1 (Payton):

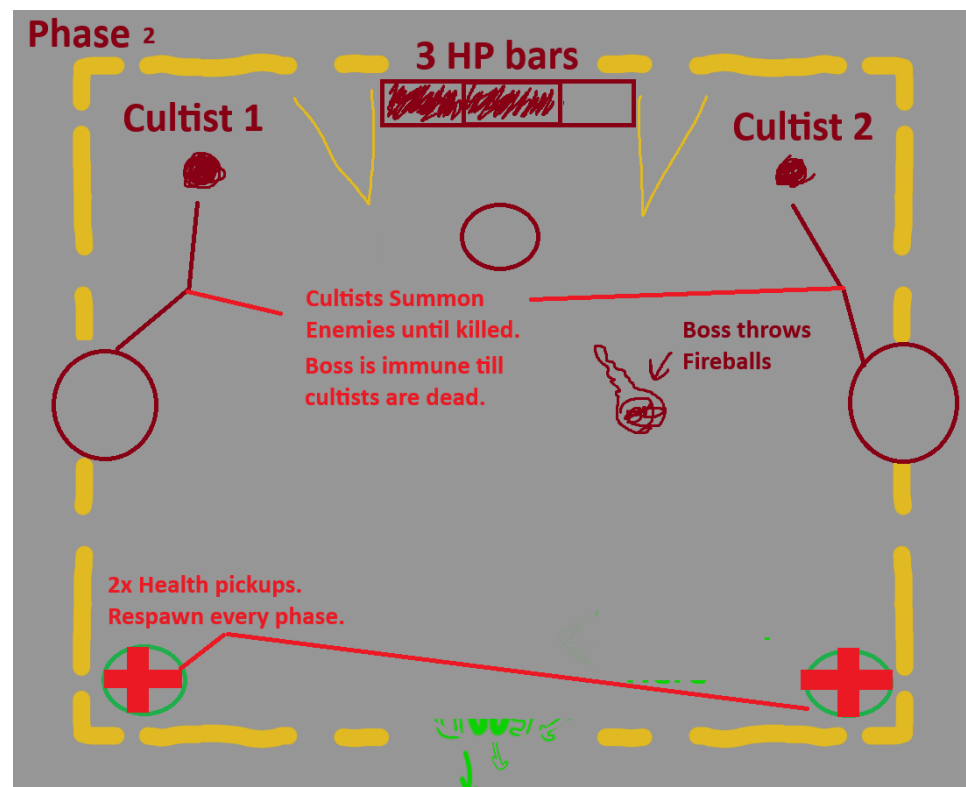
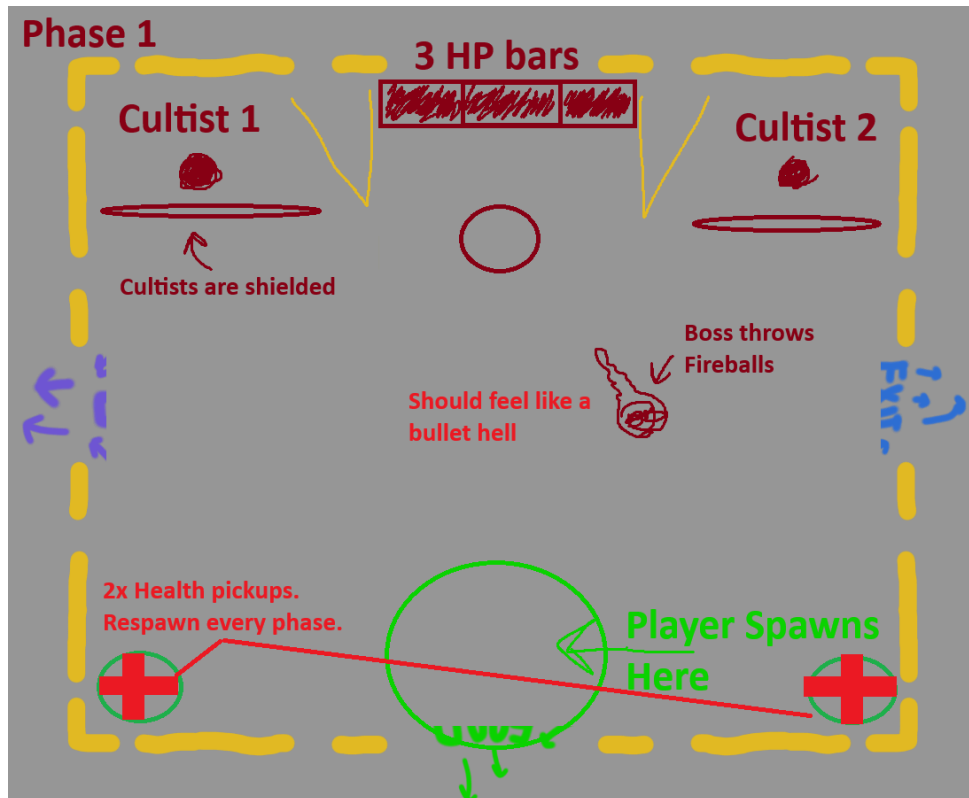


Map System Rework Proposal (Payton)

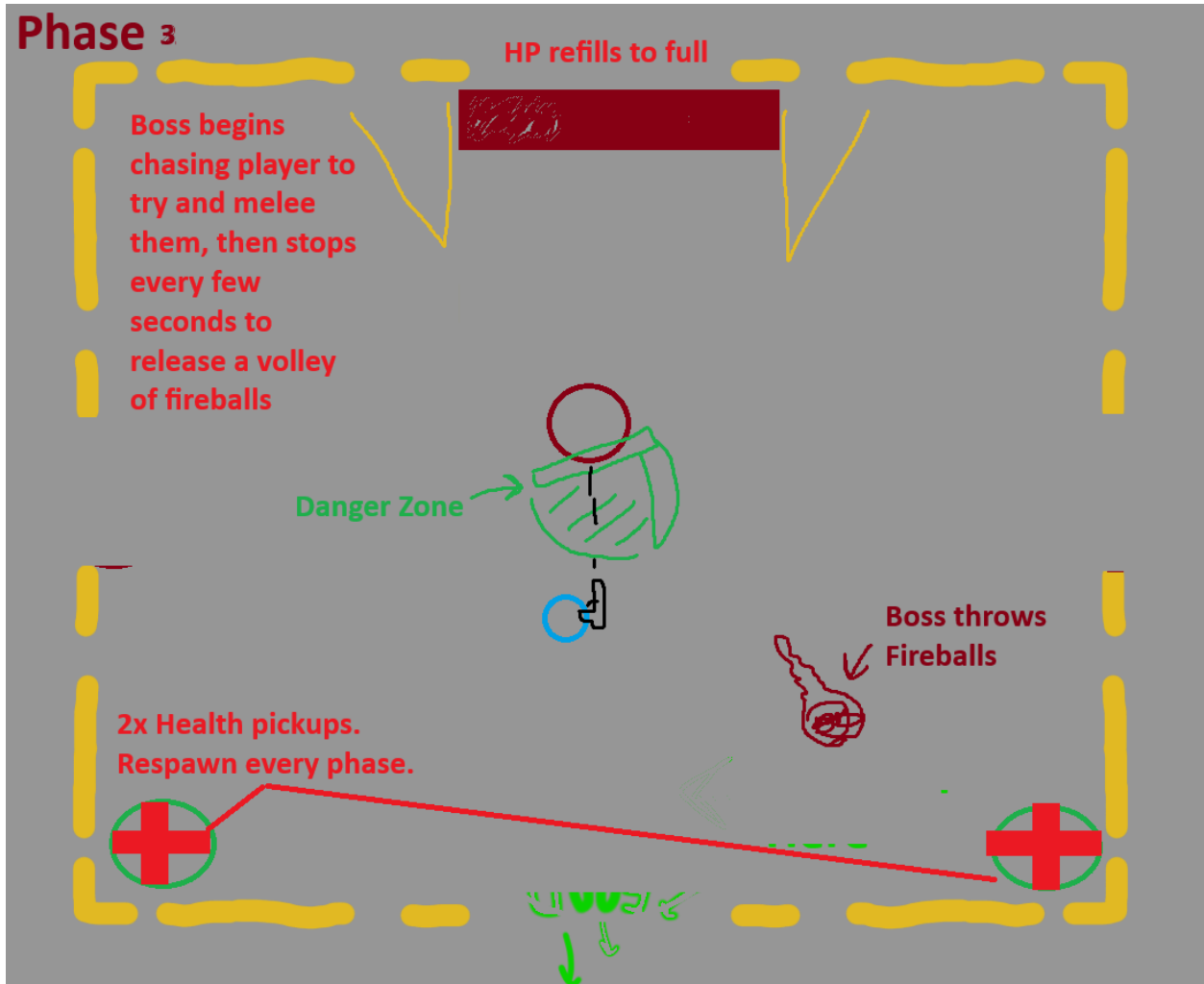
Instead of storing the neighbouring room script info in the room scripts themselves (unscalable), have the map be based on grid coordinates that increment X and Y values based on which door you take. That way, we don't need 100's of custom rooms, and can just put the map script into its own thing, and have 25 rooms that get called as needed. (I miss calculated and it was only 10 room variants).



Boss Fight Design Diagrams (Payton)



Phase 3



The bossfight (if we have time for it), is essentially focused on telling the story of the warlock as he shows off his powers of magic and undead minions, but then true to the name of the game chases you head on in anger.

Character Descriptions (Writing Style)

This section is mostly brainstorming different lines we want the characters to have.

Main villain (Evil Warlock Raph)

Basically the polar opposite of the real Raph. A hotheaded commander of the undead.

Player Character

Sort of a silent protagonist that gives a one liner about having a sore throat, then goes to play minecraft.

Narrator

Technically talks to the player directly at the end.

Text Box/Map Implementation (Payton)

I just realized I didn't actually make any notes while I was doing the text box implementation, so these are retroactive notes I made regarding how I went about doing that.

As with previous assignments, Anaha/Owen were both huge at helping in these areas. Owen walked me through Text Box implementation whenever I got stuck, and Anaha did like 90% of the difficult parts of the map, I just assembled it once we had a rudimentary fake grid system.

For the map, the grid system is based on discussions and images I exchanged with Anaha, some of which are featured above in this document. Essentially, the previous system of map generation used code given to us by Zhaoyu, which seemingly entirely lacked any ability to easily edit things as all of the rooms had to be coded into the rooms they connected to, meaning each room was its own script and was like 5-10 lines long.

I proposed a fix of a "grid" system, a vector2 that simply used fake x, y coordinates to determine where rooms were, and then had some other attached variable governing which room type to call. Anaha built this, then went further and made it a vector4 containing the aforementioned vector2, as well as room type information, and more critically the "key level" aka what level of key you needed to unlock that room. This let me build the map incredibly quickly and easily.

For the real bulk of my efforts, dialogue, the text box, and the custom font, I had help creating what is essentially just really big string arrays that are formatted to look more like specific text lines being called.

The font implementation was surprisingly simple, I just downloaded the bitcount font because it looked vaguely retro, Owen helped me call it from a folder within the project file, then I just adjusted the kerning via the way the api said to write the command, increasing it to 2 pixels (the default kerning made it near unreadable). I also made a text box by adding two squares on top of each other that are called via the textbox function whenever one of the dialogue arrays are called.

Misc (Payton)

A lot of other contributions I made were to small things that mostly involved tweaking dialogue related aspects of other scripts like the text for picking up items or opening doors.

As mentioned, I handled the bulk of the Map design/implementation after Anaha built all the foundation for me to do it. Where rooms were, where enemies were inside those rooms. I'd hardly call this the hard part since it was mostly copy pasting then tweaking numbers, but I did it so I should mention it here.

Process Work (Anaha)

Timer

SE

"

seconds EIDPsec can be used to make a
Timer by

$SE + \text{Timer} = TS$ if $SE \geq TS$
trigger something

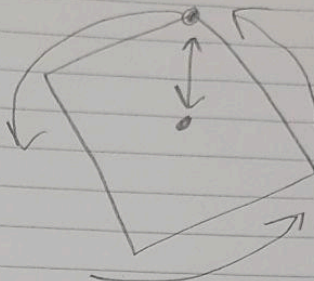
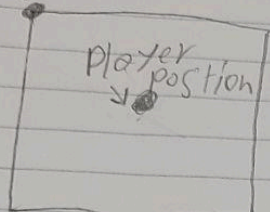
TL can be an array $TS = 0$

public virtual bool Timer (int TimerIndex, float -
Time)

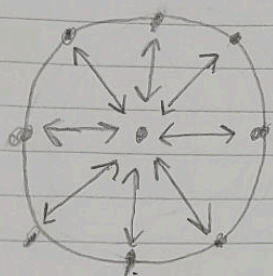
$TS.\text{Find}[\text{TimerIndex}]$

WUO
10/10/2020

Image
Start



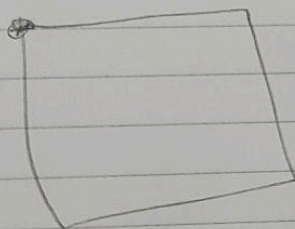
Maintain distance
from Player position and
Rotate around it



Rotate a vector around
a point

Based on

Player Rotation

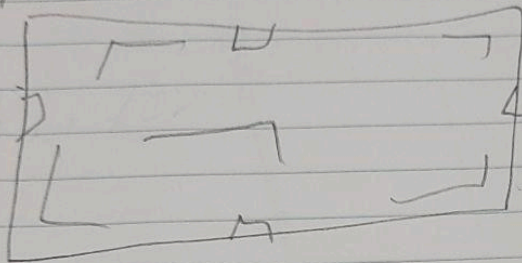


No

Character

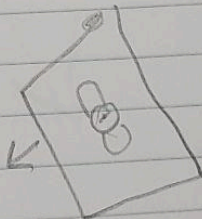
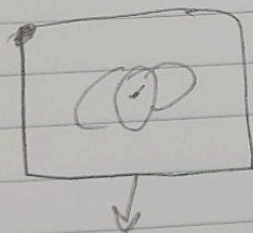
Top down, Lenger of Zelda,
Survive, Set in mohawk College,
Zombies

map



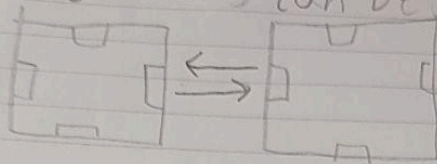
Player 1

Player 2

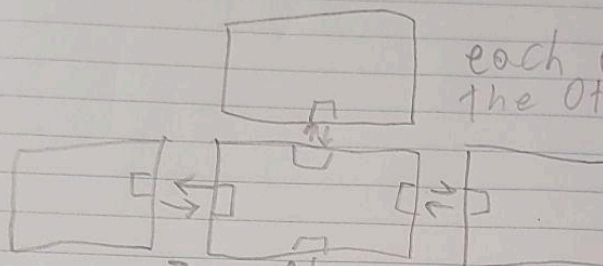


Room

Doors can be used to move between Rooms

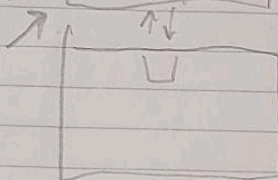


each Room has a ref. to the other Rooms around it



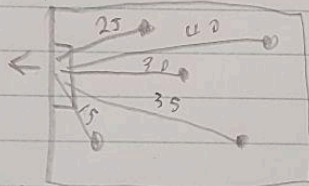
This is for Getting a ref to the next Door

Current Room



Enemies can move between Rooms

When the Player moves into a new Room



All Enemies that can will Get the Distance to the Door that the Player last used

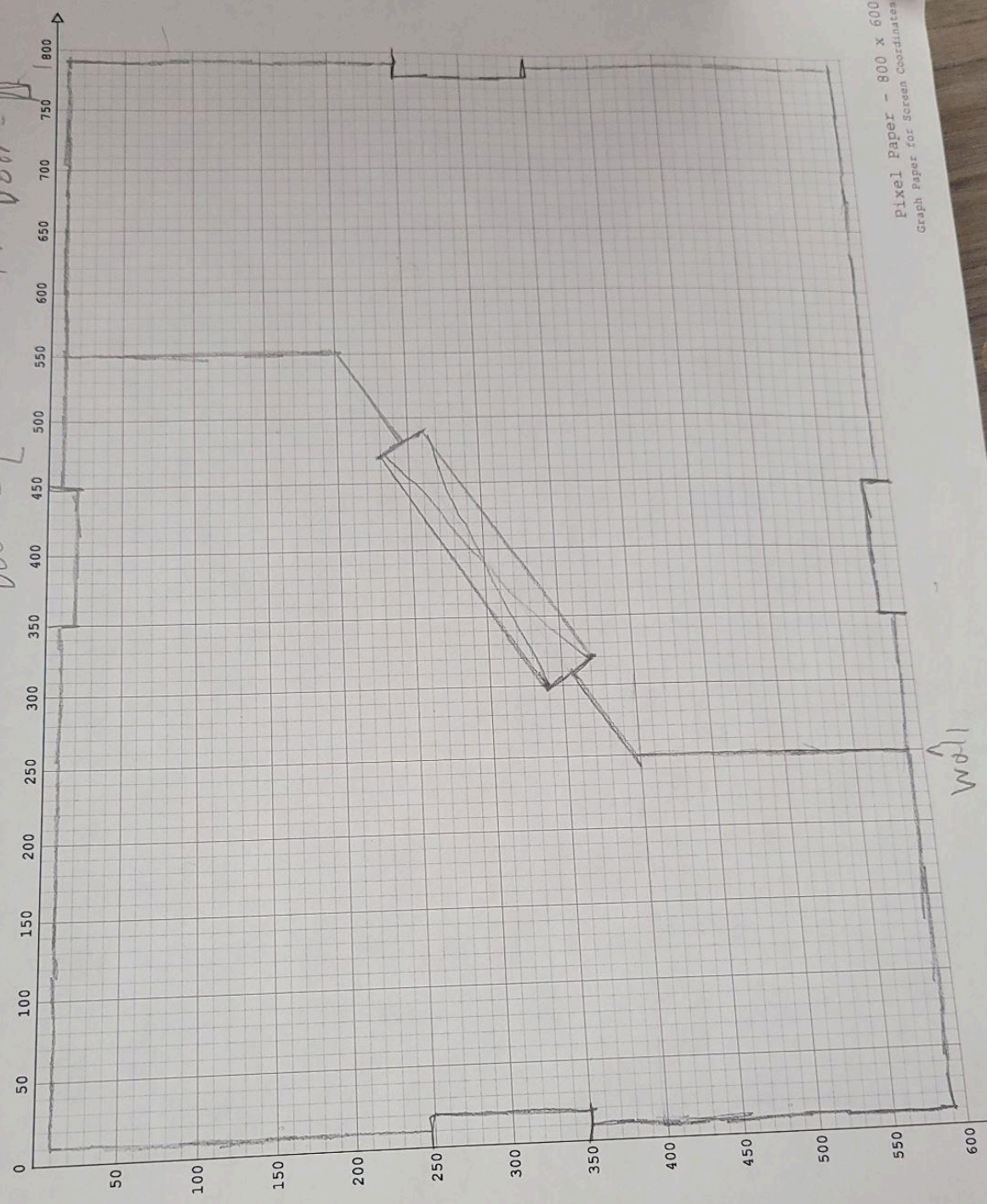
$$\left. \begin{array}{l} \text{Distance} \\ \text{Speed} \end{array} \right\} = \text{Time}$$

based on the Distance Set a timer. when the timer ends the Enemy enters the new Room.

basic Room

Door = [

puzzle Door =



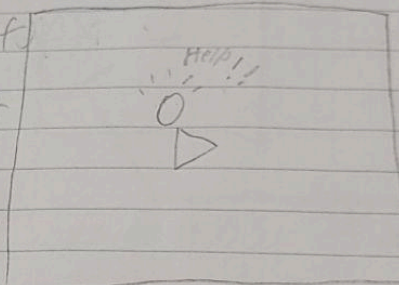
pixel paper - 800 x 600
Graph paper for Screen Coordinates

Enemies
Goal: find and kill the closest player

Zombies

Zombie player
△ ○

When in range of
their target they
slowdown the target
and themselves.



After being in range
of their target for
Num of seconds damage
the target

Damage = 00000

Move Speed = 00000

Health = 00000

fast Zombies

Just a Z@mbie but fast, less Health, more damage

Damage = 00000

Move Speed = 00000

Health = 00000

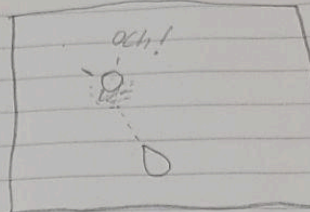
Spitters

they try to stay away from their target and shoot spit at their target.

they try to get behind their target

if the target gets too close they will act like a zombie

Spitter Player



Damage = ●●●●●
MoveSpeed = ●●●●●
Health = ●●●●●

Tanks

a fat Zombie that can take a lot of damage.

can't be stunned or pushed.

Damage = ●●●●●
MoveSpeed = ●●●●●
Health = ●●●●●

Rooms

Vector4(vector2(x,y), 5, 2)

Room Grid Position Room Type Room Code

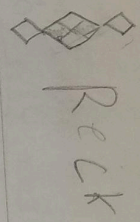
Room types

- 0. T
- 1. T
- 2. L
- 3. T
- 4. T
- 5. T
- 6. T
- 7. T
- 8. T
- 9. T
- 10. T

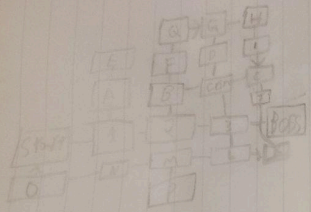
Room code

if (RC > 0)
Then Room = Locked;
if (Key code == Room code)
Then Room = unlocked;
Room.w = 0;

Amro



Rock



Door (Num)

Room array of Doors

0 Game Get Close Room (col, Row)
1 Return Close Room

col

0,0	1,0	2,0	3,0	4,0
0,1	1,1	2,1	3,1	4,1
0,2	1,2	2,2	3,2	4,2
0,3	1,3	2,3	3,3	4,3

max size

4

